Analytic Topology Miniproject

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1. Introduction

This project studies σ-topologies as structures of observable predicates closed under finite conjunction and countable disjunction. The motivating idea comes from computability over continuous data: finite information about a point typically determines not a single state, but a neighborhood of compatible states. In standard examples, semidecidable predicates are therefore open.

The aim of the project is to isolate the countable positive fragment of this phenomenon. First I recall the basic example of semidecidability on the real line. I then abstract the relevant closure properties into the notion of a σ-topology, explain its logical semantics via Sierpiński space, and finally introduce computable countably presented σ-spaces, where effective σ-openness corresponds to positive semidecidability.

The computable-topological background for semidecidability and effective openness comes from work on effective and computable topological spaces [1, 2]. The logical reading via Sierpiński space and observable predicates is closely related both to synthetic topology [3] and to the point-free, logic-oriented treatment of topology in [5]. For the broader framework of represented spaces, see [4].

2. The real line as a prototype of positive observation

The motivating example comes from computation over the real line. The basic phenomenon is that a machine does not access a real number directly, but only through progressively better approximations. Finite information about a real therefore determines not a single point, but a neighborhood of compatible points. This is the reason semidecidable predicates turn out to be open.

By an *effective coding* of the rationals, I mean a fixed surjective map

$$\nu_{\mathbb{Q}} : \mathbb{N} \to \mathbb{Q}$$

such that, given $n \in \mathbb{N}$, one can compute the rational $\nu_{\mathbb{Q}}(n)$, and conversely every rational number has at least one code. Concretely, one may code a rational by a pair of integers representing its numerator and denominator.

Fix an effective coding of the rationals by natural numbers.

Definition 2.1 (Fast Cauchy name)**.** A *fast Cauchy name* of a real number $x \in \mathbb{R}$ is a sequence of rationals $(q_n)_{n\in\mathbb{N}}$ such that

$$|x - q_n| \leq 2^{-n} \qquad \text{for every } n \in \mathbb{N}.$$

Thus the real number is presented to a machine not as a completed object, but as an infinite stream of rational approximations.

A set $W \subseteq \mathbb{N}$ is called *computably enumerable* (c.e.) if there is a Turing machine which lists exactly the elements of W, possibly with repetition and possibly without halting. Equivalently, W is the domain or range of a partial computable function.

Definition 2.2 (Effectively open subset of $\mathbb{R}$)**.** A set $U \subseteq \mathbb{R}$ is *effectively open* if there is a c.e. set $W \subseteq \mathbb{N}$ coding rational open intervals (a_i, b_i) such that

$$U = \bigcup_{i \in W} (a_i, b_i).$$

Definition 2.3 (Semidecidability on $\mathbb{R}$)**.** A set $A \subseteq \mathbb{R}$ is *semidecidable* if there is a Turing machine M such that for every fast Cauchy name p of a real x:

- if $x \in A$, then $M(p)$ halts and returns 1;
- if $x \notin A$, then $M(p)$ either does not halt or halts and returns 0.

The key point is that if the machine halts, then it has done so after reading only finitely many terms of the input name. So successful semidecision is based on finite positive evidence.

To see this concretely, first consider a rational open interval (a, b) with $a < b$. Suppose that (q_n) is a fast Cauchy name of a real x. Then membership in (a, b) is semidecidable by searching for some n such that

$$a + 2^{-n} < q_n < b - 2^{-n}.$$

Indeed, if this inequality holds, then

$$a < q_n - 2^{-n} \leq x \leq q_n + 2^{-n} < b,$$

so $x \in (a, b)$.

Conversely, if $x \in (a, b)$, let

$$\varepsilon = \min(x - a, b - x) > 0.$$

For sufficiently large n, one has $2^{-n} < \varepsilon/2$ and $|x - q_n| \leq 2^{-n}$, hence

$$a + 2^{-n} < q_n < b - 2^{-n}.$$

So the search eventually succeeds. Therefore rational open intervals are semidecidable.

The same argument extends from a single interval to any effectively open set.

Lemma 2.1. *Every effectively open subset of $\mathbb{R}$ is semidecidable.*

Proof. Let

$$U = \bigcup_{i \in W} (a_i, b_i)$$

where W is c.e. and each (a_i, b_i) is a rational open interval. Given a fast Cauchy name of x, run in parallel the semidecision procedures for the intervals (a_i, b_i) as their indices i are enumerated into W. Halt as soon as one of these procedures succeeds.

If $x \in U$, then $x \in (a_i, b_i)$ for some $i \in W$, and the corresponding procedure eventually halts. If $x \notin U$, then no interval procedure succeeds, so the global procedure never halts. □

The converse direction explains why openness is forced by semidecidability.

Lemma 2.2. *Every semidecidable subset of $\mathbb{R}$ is open.*

Proof. Let $A \subseteq \mathbb{R}$ be semidecidable, and let $x \in A$. Since $x \in A$, the semidecision procedure halts on every fast Cauchy name of x. Choose one such name

$$p = (q_0, q_1, q_2, \dots)$$

satisfying the stronger condition

$$|x - q_n| < 2^{-n} \qquad \text{for every } n \in \mathbb{N}.$$

Since the computation halts in finite time, it inspects only a finite prefix

$$(q_0, \dots, q_n).$$

Consider the set of reals compatible with this finite information:

$$J_n = \bigcap_{k \le n} (q_k - 2^{-k}, q_k + 2^{-k}).$$

Because $|x - q_k| < 2^{-k}$ for every $k \le n$, the point x belongs to J_n. Moreover, J_n is open.

Now let $y \in J_n$. Then for every $k \le n$ one has

$$|y - q_k| < 2^{-k},$$

so the same finite prefix $(q_0, \dots, q_n)$ is compatible with y. Hence this prefix can be extended to a fast Cauchy name of y. On that extended name, the machine performs exactly the same finite computation as on p, and therefore halts. Thus $y \in A$.

So J_n is an open neighborhood of x contained in A. Since $x \in A$ was arbitrary, the set A is open. □

Combining the two directions, one obtains the basic topological consequence of semidecidability on the real line.

Theorem 2.3. *Every effectively open subset of $\mathbb{R}$ is semidecidable, and every semidecidable subset of $\mathbb{R}$ is open.*

The real line already exhibits the pattern that will be abstracted later. A finite amount of information about a point determines a neighborhood of compatible points rather than a single point. A successful semidecision therefore depends on finitely many positive observations, while an effectively open predicate is obtained by countable search through possible such certificates.

This is exactly the closure pattern isolated by a σ-topology: finite conjunction of observable conditions and countable disjunction of possible witnesses. The later sections recast this structure in an abstract form.

3. σ-topologies and strict examples

Motivated by the previous discussion of semidecidability on $\mathbb{R}$, I now isolate the closure pattern used there. The relevant operations were finite conjunction of observable conditions and countable disjunction of possible witnesses. This leads to the following weakening of ordinary topology.

Definition 3.1 (σ-topology)**.** Let X be a set. A family $\Sigma \subseteq \mathcal{P}(X)$ is called a **σ-topology** on X if:

- $\varnothing, X \in \Sigma$;
- whenever $U, V \in \Sigma$, then $U \cap V \in \Sigma$;
- whenever $(U_n)_{n \in \mathbb{N}} \subseteq \Sigma$, then $\bigcup_{n \in \mathbb{N}} U_n \in \Sigma$.

The members of Σ are called **σ-open** sets. A pair (X, Σ) is called a **σ-topological space**.

Every ordinary topology is a σ-topology, since closure under arbitrary unions implies closure under countable unions. The interest of the definition is that the converse fails: a σ-topology need not be a topology.

Definition 3.2 (Strict σ-topology)**.** A σ-topology which is not closed under arbitrary unions will be called a **strict σ-topology**.

The simplest example is obtained by forcing openness to remain countable.
Let X be an uncountable set and define

$$\Sigma_\omega(X) = \{A \subseteq X : A \text{ is countable}\} \cup \{X\}.$$

Then $\Sigma_\omega(X)$ is a strict σ-topology on X.

Proof. Clearly $\varnothing \in \Sigma_\omega(X)$ and $X \in \Sigma_\omega(X)$. Finite intersections of countable sets are countable, and the union of countably many countable sets is countable. Of course unions of opens including X are equal to X, but they are still in $\Sigma_\omega(X)$ by construction. Thus $\Sigma_\omega(X)$ is a σ-topology.
However, it is not a topology. Indeed, if $A \subseteq X$ is any uncountable proper subset, then

$$A = \bigcup_{x \in A} \{x\},$$

and each singleton $\{x\}$ is countable, hence σ-open. But A itself is neither countable nor equal to X, so $A \notin \Sigma_\omega(X)$. Thus closure under arbitrary unions fails. □

This example already shows that a strict σ-topology behaves differently from an ordinary topology not only set-theoretically but also logically. In an ordinary topological space, the interior of a set is obtained as the union of all open subsets contained in it. In a strict σ-topology, that union may require uncountably many σ-open pieces and therefore need not itself be σ-open.

3.1 Pointwise interior and closure

The failure of arbitrary unions means that the usual interior and closure constructions need not exist internally in a σ-topology.

Take the space $(X, \Sigma_\omega(X))$ above, and let $A \subseteq X$ be an uncountable proper subset. Every singleton $\{x\}$ with $x \in A$ is σ-open and satisfies $\{x\} \subseteq A$. Hence every point of A lies in some σ-open subset of A. Set-theoretically,

$$A = \bigcup \{U \in \Sigma_\omega(X) : U \subseteq A\},$$

but A itself is not σ-open. So the ordinary formula for interior does not define an internal operation on arbitrary subsets.

The dual phenomenon occurs for closure. If $A \subseteq X$ is countable and nonempty, then every σ-open neighborhood of a point of A meets A, while for each $x \notin A$ the singleton $\{x\}$ is a σ-open neighborhood disjoint from A. Thus the points adherent to A are exactly the points of A itself. But A is not σ-closed, since its complement is an uncountable proper subset and hence not σ-open.

This suggests introducing pointwise notions of interior and closure without requiring them to land back inside Σ.

Definition 3.3 (Core)**.** Let (X, Σ) be a σ-space and let $A \subseteq X$. The **σ-core** of A is

$$\mathrm{cor}_\sigma(A) = \{x \in X : \exists U \in \Sigma \ (x \in U \subseteq A)\}.$$

Definition 3.4 (Adherence)**.** Let (X, Σ) be a σ-space and let $A \subseteq X$. The **σ-adherence** of A is

$$\mathrm{adh}_\sigma(A) = \{x \in X : \forall U \in \Sigma\,(x \in U \Rightarrow U \cap A \neq \varnothing)\}.$$

These two operators satisfy the expected De Morgan duality:

$$\mathrm{adh}_\sigma(A) = X \setminus \mathrm{cor}_\sigma(X \setminus A),$$

$$\mathrm{cor}_\sigma(A) = X \setminus \mathrm{adh}_\sigma(X \setminus A).$$

The point is that in a strict σ-topology, $\mathrm{cor}_\sigma(A)$ need not be σ-open and $\mathrm{adh}_\sigma(A)$ need not be σ-closed. Thus these are externally defined pointwise operations, not in general internal lattice operations on the family of σ-open sets.

3.2 σ-continuity

The appropriate notion of morphism is the evident one.

Definition 3.5 (σ-continuity)**.** Let (X, Σ_X) and (Y, Σ_Y) be σ-topological spaces.
A map $f : X \to Y$ is **σ-continuous** if for every $V \in \Sigma_Y$ one has

$$f^{-1}(V) \in \Sigma_X.$$

Thus σ-topological spaces form a category analogous to the ordinary category of topological spaces. In the following section, the family of σ-open sets will be reinterpreted algebraically as a system of observable predicates closed under finite conjunction and countable disjunction.

4. σ-frames, observable predicates, and Sierpiński semantics

The correct algebraic object underlying a σ-topology is not merely a family of subsets, but a system of predicates closed under finite conjunction and countable disjunction. This leads naturally to the following definition.

Definition 4.1 (σ-frame)**.** A **σ-frame** is a partially ordered set L with:

- finite meets $a \wedge b$ and top element $\top$,
- countable joins $\bigvee_{n\in\mathbb{N}} a_n$ and bottom element $\bot$,

such that finite meets distribute over countable joins:

$$a \wedge \bigvee_{n\in\mathbb{N}} b_n = \bigvee_{n\in\mathbb{N}} (a \wedge b_n).$$

This is the countable analogue of a frame from ordinary topology. In the present setting, countable joins correspond to countable disjunctions of observable properties.

Let (X, Σ) be a σ-topological space. Then the ordered set $(\Sigma, \subseteq)$ is a σ-frame, where:

$$\bot = \varnothing, \qquad \top = X,$$

finite meets are given by finite intersections, and countable joins are given by countable unions.

Proof. By definition, Σ contains $\varnothing$ and X, is closed under finite intersections, and is closed under countable unions. The distributive law

$$U \cap \bigcup_n V_n = \bigcup_n (U \cap V_n)$$

holds in the powerset algebra $\mathcal{P}(X)$, hence also in Σ. □

Thus a σ-topology may be viewed as a σ-frame of observable predicates on X. This shifts attention from points themselves to the properties of points that can be recognized by finite positive evidence and countable search.

4.1 Sierpiński space and observable predicates

The ordinary Sierpiński space is the two-point space

$$\mathbb{S} = \{\bot, \top\}$$

with open sets

$$\varnothing, \qquad \{\top\}, \qquad \mathbb{S}.$$

Since this family is also closed under countable unions, $\mathbb{S}$ is likewise a σ-topological space.

Its importance is that σ-open predicates on a space X correspond exactly to σ-continuous maps into $\mathbb{S}$.

Let (X, Σ) be a σ-topological space and let $A \subseteq X$. Then the following are equivalent:

1. $A \in \Sigma$;
2. the characteristic map

$$\chi_A : X \to \mathbb{S}, \qquad \chi_A(x) = \begin{cases} \top & x \in A, \\ \bot & x \notin A \end{cases}$$

is σ-continuous.

Proof. The only nontrivial σ-open subset of $\mathbb{S}$ is $\{\top\}$. Therefore χ_A is σ-continuous iff

$$\chi_A^{-1}(\{\top\}) = A$$

is σ-open. □

So it is natural to write

$$\mathcal{O}_\sigma(X) := C_\sigma(X, \mathbb{S})$$

for the collection of all σ-open predicates on X.

Remark. The set $\mathcal{O}_\sigma(X)$ may be identified with Σ, but the notation $C_\sigma(X, \mathbb{S})$ emphasizes the logical reading: an open set is not merely a subset of X, but an observable predicate with values in the space of semidecision outcomes. This viewpoint is closely related to the synthetic use of Sierpiński space as a space of semidecision outcomes and observable predicates [3], and more broadly to the logical treatment of topology developed in [5].

4.2 Positive logic on σ-spaces

The Sierpiński space carries the basic positive logical operations

$$\wedge, \vee : \mathbb{S} \times \mathbb{S} \to \mathbb{S}$$

given by the usual truth tables, together with countable disjunction

$$\bigvee : \mathbb{S}^{\mathbb{N}} \to \mathbb{S}, \qquad \bigvee((s_n)_n) = \top \iff \exists n\ s_n = \top.$$

These operations induce the corresponding connectives on observable predicates.

Let $p, q : X \to \mathbb{S}$ be σ-continuous, and let $(p_n)_{n\in\mathbb{N}}$ be a countable family of σ-continuous maps. Then the predicates

$$p \wedge q, \qquad p \vee q, \qquad \bigvee_{n\in\mathbb{N}} p_n$$

are again σ-continuous. Moreover,

$$(p \wedge q)^{-1}(\{\top\}) = p^{-1}(\{\top\}) \cap q^{-1}(\{\top\}),$$
$$(p \vee q)^{-1}(\{\top\}) = p^{-1}(\{\top\}) \cup q^{-1}(\{\top\}),$$
$$\left(\bigvee_n p_n\right)^{-1}(\{\top\}) = \bigcup_n p_n^{-1}(\{\top\}).$$

Proof. Each statement follows immediately by taking inverse images of $\{\top\}$ and using closure of Σ under finite intersections and countable unions. □

Hence $\mathcal{O}_\sigma(X)$ interprets a propositional logic with:

$$\bot, \quad \top, \quad \wedge, \quad \vee, \quad \bigvee_{n\in\mathbb{N}}.$$

This is precisely the logic generated by finite conjunction and countable disjunction of observable predicates.

4.3 Why negation and implication fail in general

The essential point is that a general strict σ-topology canonically supports only positive countable logic. Finite conjunction and countable disjunction are always available, but implication and negation are not available in general.

Negation

$$\neg : \mathbb{S} \to \mathbb{S}$$

is not σ-continuous, because

$$\neg^{-1}(\{\top\}) = \{\bot\},$$

and $\{\bot\}$ is not σ-open in $\mathbb{S}$.

This reflects the computational asymmetry of semidecidability: positive information is stable under finite observation, while negative information is not.

In an ordinary topological space, the open sets form a frame, hence in particular a Heyting algebra. Thus implication and negation are available internally, for example by

$$U \Rightarrow V = \mathrm{Int}((X \setminus U) \cup V), \qquad \neg U = \mathrm{Int}(X \setminus U).$$

The key reason this need not persist for a strict σ-topology is that Heyting implication requires the existence of the largest σ-open set W such that

$$W \cap U \subseteq V.$$

In an ordinary topology this largest open exists because one may take the union of all open sets satisfying this condition, and arbitrary unions of open sets are again open.

By contrast, in a strict σ-topology one has closure only under countable unions. The union of all σ-open sets W with

$$W \cap U \subseteq V$$

may therefore fail to be σ-open, since that union may be uncountable. Consequently, the required largest W need not exist, and so implication need not be available. The same applies in particular to negation, which is the special case

$$\neg U = (U \Rightarrow \varnothing).$$

Thus the passage from topology to σ-topology is not only a weakening of closure under unions. It is also a weakening of logical strength:

- ordinary topologies support intuitionistic logic through the Heyting algebra of open sets
- σ-topologies always support the positive fragment generated by finite conjunction and countable disjunction
- implication and negation are not guaranteed in general, and require additional structure beyond the axioms of a σ-topology.

4.4 Functoriality

The algebraic viewpoint also clarifies the role of σ-continuous maps.

If $f : (X, \Sigma_X) \to (Y, \Sigma_Y)$ is σ-continuous, then the inverse image map

$$f^{-1} : \Sigma_Y \to \Sigma_X$$

preserves finite meets and countable joins. Equivalently,

$$f^{-1} : \mathcal{O}_\sigma(Y) \to \mathcal{O}_\sigma(X)$$

is a homomorphism of σ-frames.

Proof. For subsets of Y, inverse image always preserves finite intersections and arbitrary unions:

$$f^{-1}(U \cap V) = f^{-1}(U) \cap f^{-1}(V), \qquad f^{-1}\left(\bigcup_n U_n\right) = \bigcup_n f^{-1}(U_n).$$

Since f is σ-continuous, these inverse images lie in Σ_X whenever the original sets lie in Σ_Y. □

This is the precise sense in which a σ-space carries a logic of observable predicates: points are secondary, while σ-continuous maps act contravariantly on the corresponding σ-frames of observations.

5. Computable countably presented σ-spaces

So far the discussion has been purely extensional: a σ-space carries a family of observable predicates closed under finite conjunction and countable disjunction. To obtain genuine computational content, however, one must specify a *countable* stock of primitive observations from which all others are generated. This is the role of a σ-basis.

Definition 5.1 (σ-basis)**.** Let (X, Σ) be a σ-topological space. A family

$$\mathcal{B} \subseteq \Sigma$$

is called a **σ-basis** if every σ-open set is a countable union of members of $\mathcal{B}$, that is,

$$\forall U \in \Sigma \; \exists (B_n)_{n\in\mathbb{N}} \subseteq \mathcal{B} \quad U = \bigcup_{n\in\mathbb{N}} B_n.$$

The usual local basis condition from ordinary topology,

$$\forall x \in U \in \Sigma \; \exists B \in \mathcal{B} \; (x \in B \subseteq U),$$

is no longer sufficient by itself. It shows only that every point of U lies in some basic set contained in U, but the family of all such basic sets may be uncountable. Since a σ-topology is closed only under countable unions, one needs an additional countability condition.

Lemma 5.1. *Let (X, Σ) be a σ-space and let $\mathcal{B} \subseteq \Sigma$. Assume:*

1. *for every $x \in U \in \Sigma$ there exists $B \in \mathcal{B}$ such that $x \in B \subseteq U$;*
2. *every $\mathcal{B}$-cover of a σ-open set admits a countable subcover.*

Then $\mathcal{B}$ is a σ-basis.

Proof. Let $U \in \Sigma$. By (1), for every $x \in U$ there exists $B_x \in \mathcal{B}$ such that

$$x \in B_x \subseteq U.$$

Hence

$$U = \bigcup_{x \in U} B_x.$$

This is a $\mathcal{B}$-cover of U, so by (2) it has a countable subcover

$$U = \bigcup_{n \in \mathbb{N}} B_n$$

with each $B_n \in \mathcal{B}$. Therefore $\mathcal{B}$ is a σ-basis. □

Remark (σ-subbasis). A family

$$\mathcal{S} \subseteq \mathcal{P}(X)$$

may be called a **σ-subbasis** if finite intersections of elements of $\mathcal{S}$ form a σ-basis. In computational examples, the members of $\mathcal{S}$ can be thought of as primitive tests, while finite intersections represent finite conjunctions of observations.

The computational role of a countable σ-basis is that its members are primitive semidecidable observations. A point is then described by listing which basic observations hold of it.

The general background here comes from effective topological spaces and computable topology [1, 2], together with the represented-spaces perspective [4].

Definition 5.2 (Countably presented σ-space)**.** A **countably presented σ-space** is a pair

$$(X, (B_n)_{n \in \mathbb{N}})$$

such that $(B_n)_{n \in \mathbb{N}}$ is a countable σ-basis for a σ-topology Σ_X on X.

The following simple lemma provides the basic limitation of contable presentability:

Lemma 5.2. *If X is uncountable and every singleton $\{x\}$ is σ-open, then X cannot have a countable σ-basis.*

Proof: If $\{x\} = \bigcup_n B_n$ with each B_n from a basis, then every nonempty B_n must be contained in x, hence equal to x. So each singleton must itself be a basic element. That forces uncountably many basis elements. □

To recover points from their observable properties, one needs the usual separation assumption.

Definition 5.3 (T_0)**.** A σ-space (X, Σ) is T_0 if for any distinct points $x \neq y$ there exists $U \in \Sigma$ such that either

$$x \in U,\ y \notin U \qquad \text{or} \qquad y \in U,\ x \notin U.$$

If X is T_0, then the family of basic opens containing a point determines that point uniquely. This leads to the positive-information representation.

Definition 5.4 (Neighborhood enumeration)**.** Let $(X, (B_n))$ be a countably presented T_0 σ-space. For $x \in X$, define

$$N(x) := \{n \in \mathbb{N} : x \in B_n\}.$$

A **name** of x is any infinite sequence enumerating the set $N(x)$, possibly with repetitions.

Thus a name of a point does not describe the point directly, but rather enumerates the primitive observations true of that point.

To make the resulting notion of observability effective, one must be able to compute finite conjunctions of basic observations.

Definition 5.5 (Computable σ-space)**.** A countably presented T_0 σ-space is **computable** if there is a uniform effective procedure which, given $i, j \in \mathbb{N}$, enumerates a c.e. set

$$W_{i,j} \subseteq \mathbb{N}$$

such that

$$B_i \cap B_j = \bigcup_{n \in W_{i,j}} B_n.$$

By iterating this condition, finite intersections of basic opens are uniformly effectively σ-open. This is exactly what is needed to model finite positive evidence.

Definition 5.6 (Effectively σ-open set)**.** Let X be a computable countably presented σ-space. A set $U \subseteq X$ is **effectively σ-open** if

$$U = \bigcup_{n \in W} B_n$$

for some c.e. set $W \subseteq \mathbb{N}$.

Because names enumerate only positive information, the appropriate notion of semidecidability is monotone: membership must be witnessed by finitely many basic observations.

Definition 5.7 (Positive semidecidability)**.** Let X be a computable countably presented T_0 σ-space. A set $A \subseteq X$ is **positively semidecidable** if there exists a c.e. family

$$\mathcal{F} \subseteq \mathcal{P}_{\text{fin}}(\mathbb{N})$$

such that

$$x \in A \iff \exists F \in \mathcal{F}\ (F \subseteq N(x)).$$

In other words, $x \in A$ holds exactly when one of countably many finite positive certificates is contained in the observable data of x.

Lemma 5.3 (Uniform finite intersections)**.** *Let X be a computable countably presented σ-space with basis $(B_n)_{n \in \mathbb{N}}$. Then there is a uniform effective procedure which, given a finite set $F \subseteq \mathbb{N}$, enumerates a c.e. set*

$$W_F \subseteq \mathbb{N}$$

such that

$$\bigcap_{i \in F} B_i = \bigcup_{n \in W_F} B_n.$$

For $F = \varnothing$, we take

$$\bigcap_{i \in \varnothing} B_i = X.$$

Proof. We argue by induction on the size of F.

If $F = \varnothing$, then

$$\bigcap_{i \in \varnothing} B_i = X.$$

Since $(B_n)_{n\in\mathbb{N}}$ is a σ-basis, the space X is the union of all basic opens:

$$X = \bigcup_{n\in\mathbb{N}} B_n.$$

Thus we may take

$$W_\varnothing = \mathbb{N},$$

which is c.e.

If $F = \{i\}$, then

$$\bigcap_{j\in F} B_j = B_i = \bigcup_{n\in\{i\}} B_n,$$

so we may take $W_F = \{i\}$.

Now suppose that, for some finite $F \subseteq \mathbb{N}$, we have already computed a c.e. set $W_F \subseteq \mathbb{N}$ such that

$$\bigcap_{i\in F} B_i = \bigcup_{n\in W_F} B_n,$$

and let $j \in \mathbb{N}$. Then

$$\left(\bigcap_{i\in F} B_i\right) \cap B_j = \left(\bigcup_{n\in W_F} B_n\right) \cap B_j = \bigcup_{n\in W_F} (B_n \cap B_j).$$

By computability of the presentation, for each pair n, j there is a uniformly c.e. set $W_{n,j} \subseteq \mathbb{N}$ such that

$$B_n \cap B_j = \bigcup_{m\in W_{n,j}} B_m.$$

Therefore

$$\left(\bigcap_{i\in F} B_i\right) \cap B_j = \bigcup_{n\in W_F} \bigcup_{m\in W_{n,j}} B_m.$$

Since W_F is c.e. and the family $(W_{n,j})_{n,j}$ is uniformly c.e., the set

$$W_{F\cup\{j\}} := \bigcup_{n\in W_F} W_{n,j}$$

is again c.e., uniformly in F and j. Hence

$$\bigcap_{i\in F\cup\{j\}} B_i = \bigcup_{m\in W_{F\cup\{j\}}} B_m.$$

This gives the required procedure. □

Theorem 5.4. *Let X be a computable countably presented T_0 σ-space. For a subset $A \subseteq X$, the following are equivalent:*

1. *A is effectively σ-open;*
2. *A is positively semidecidable.*

Proof. Suppose first that

$$A = \bigcup_{n\in W} B_n$$

for some c.e. set $W \subseteq \mathbb{N}$. Let

$$\mathcal{F} = \big\{\{n\} : n \in W\big\} \subseteq \mathcal{P}_{\text{fin}}(\mathbb{N}).$$

Then for every $x \in X$,

$$x \in A \iff \exists n \in W\ (x \in B_n) \iff \exists F \in \mathcal{F}\ (F \subseteq N(x)).$$

So A is positively semidecidable.

Conversely, suppose that A is positively semidecidable via a c.e. family

$$\mathcal{F} \subseteq \mathcal{P}_{\text{fin}}(\mathbb{N}).$$

For each finite set $F \in \mathcal{F}$, define

$$U_F := \bigcap_{i \in F} B_i,$$

with the convention $U_\varnothing = X$. Then for every $x \in X$,

$$x \in U_F \iff F \subseteq N(x).$$

Hence

$$x \in A \iff \exists F \in \mathcal{F}\ (F \subseteq N(x)) \iff x \in \bigcup_{F \in \mathcal{F}} U_F.$$

Therefore

$$A = \bigcup_{F \in \mathcal{F}} U_F.$$

By the previous lemma, each U_F is effectively σ-open, uniformly in F: there is a uniformly c.e. set $W_F \subseteq \mathbb{N}$ such that

$$U_F = \bigcup_{n \in W_F} B_n.$$

Since $\mathcal{F}$ is c.e., the union

$$W := \bigcup_{F \in \mathcal{F}} W_F$$

is again c.e. Consequently,

$$A = \bigcup_{F \in \mathcal{F}} U_F = \bigcup_{F \in \mathcal{F}} \bigcup_{n \in W_F} B_n = \bigcup_{n \in W} B_n,$$

so A is effectively σ-open. □

Computationally, this theorem expresses the same logical structure identified earlier at the level of σ-frames: observable predicates are obtained from primitive observations by finite conjunction and countable effective disjunction.

6. Conclusions and further directions

The main point of this project has been to isolate, in abstract form, the closure pattern that appears in semidecidability over continuous data. On the real line, semidecidable predicates are governed by finite positive evidence and countable search. Abstracting this leads to the notion of a σ-topology: a family of observable predicates closed under finite intersections and countable unions.

From this point of view, a σ-topology is naturally understood as a σ-frame of observable predicates, and Sierpiński space provides the corresponding logical semantics. In the countably presented computable setting, this structure acquires an effective form: effective σ-openness corresponds exactly to positive semidecidability. Thus the abstract closure conditions match a concrete computational interpretation.

Several natural directions remain open. One is to compare the present approach more systematically with the theory of represented spaces, where semidecidable predicates are also interpreted via Sierpiński space. Another is to study more closely the algebraic side of σ-frames and the extent to which point-free methods survive when arbitrary unions are no longer available. A further question is which additional assumptions on a σ-topology allow the recovery of stronger logical structure, such as implication or a well-behaved interior operation.

Finally, it would be interesting to investigate other examples of strict σ-topologies arising in logic and verification. For instance, one may consider classes of predicates generated from closed sets by countable unions, such as F_σ-sets on Cantor space, as a possible way of encoding countable disjunctions of safety-type conditions. This suggests that σ-topologies may provide a useful framework not only for positive semidecidability, but more generally for logics of countably observable properties.

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